higher gain, at the cost of increased noise and distortion. If the amount of positive feedback is large enough, the result is oscillation where the amplifier circuit produces its own signal.

7.4.2. Using Positive Feedback

When an amplifier is operating without feedback is called as "open-loop" mode and with feedback *(either +ve or -ve)*, is known as "closed-loop" mode. In ordinary amplifiers, negative feedback is used to provide advantages in bandwidth, distortion and noise generation, and in these circuits the closed-loop gain of the amplifier is much less than the open-loop gain. However, when positive feedback is used in an amplifier system, the closed-loop gain (with feedback) will be greater than the open-loop gain; the amplifier gain is now increased by the feedback. Additional effect of positive feedback are reduced bandwidth, (but this does not a matter in an oscillator producing a sine wave having a single frequency), and increased distortion. However, even severe distortion in the amplifier is allowed in some sine wave oscillator designs, where it does not affect the shape of the output waveform.

In oscillators using positive feedback, it is important that amplitude of the oscillator output remains stable. Therefore, the closed loop gain must be 1 (unity). In other words, the gain within the loop (provided by the amplifier) should exactly match the losses (caused by the feedback circuit) within the loop. In this way, there will be no increase or decrease in the amplitude of the output signal as shown in the waveform 7.5.

This is achieved by Barkhausen criteria given by $|A\beta| = 1$.

where A is the amplifier gain and β is the transfer function of feedback.

7.4.3. The Condition for Oscillation

Positive feedback must occur at a frequency where the voltage gain of the amplifier is equal to the losses (attenuation) occurring in the feedback path. For example, 1/30 of the output signal feedback to be in phase with the input at a particular frequency, and the gain of the amplifier (with feedback) is 30 times or more, thus oscillation should take place. The conditions for oscillation are, (i) the oscillations should take place at one particular frequency and (ii) the amplified output of the oscillation should be constant.

There are many different oscillator designs in use, each design achieving the above criteria in different ways; some designers are particularly suited to producing certain wave shapes or work best within certain band of frequencies. Whatever design is used, the way of achieving a signal of constant frequency and constant amplitude, by using one or more of the following three basic methods.

Method 1: Make sure that positive feedback occurs only at one frequency of oscillation. This may be achieved by ensuring that only signal of the required frequency are feedback or by ensuring feedback signal is the correct in-phase at only one frequency.

Method 2: Make sure that sufficient amplification for oscillation can take place only the required frequency by using an amplifier that has an extremely narrow

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